Technical document
Hardware requirements

Aiuda platform

Technical specification for deployment on a self-hosted server

This document outlines the minimum, recommended, and optimal hardware requirements for deploying the Aiuda platform on a self-hosted server or virtual machine (VM). It includes estimates for storage and networking, as well as approximate costs..

1. Hardware requirements

The artificial intelligence models used by Aiuda (Whisper for transcription and NLLB for translation) require sufficient resources to operate in memory simultaneously. Three configuration levels are detailed below..

Component	Minimum	Recommended	Optimal (with GPU)
CPU	4 cores	16–24 cores	16+ cores + GPU NVIDIA
RAM	8 GB	12–16 GB	16–32 GB
GPU	No requerida	No requerida	NVIDIA CUDA, 4 GB VRAM mín.
Operative System	Ubuntu 22.04+	Ubuntu 24.04	Ubuntu 24.04
Docker	Engine 24+ / Compose v2	Engine 24+ / Compose v2	Engine 24+ / Compose v2

- With the minimum configuration (4 cores, 8 GB RAM), the platform functions, but processing may be slow. A 1-minute video can take between 2 and 4 minutes to process.
- With an NVIDIA GPU that supports CUDA, processing time is reduced by a factor of 5 to 10 compared to using the CPU.
- Together, the Whisper and NLLB models occupy approximately 4–5 GB of RAM during processing.

2. Storage

Storage is divided into two parts: fixed space for the system and AI models, and variable space for the `jobs/` folder, which grows with each task processed.

	Estimated size	Notes
Operative System + Docker	~5 GB	Fixed, does not change
IA Models (Whisper + NLLB)	~3 GB	Downloaded the first time
Docker image backend	~7 GB	Fixed after the initial build
jobs/ — video of 1h, 1 language	~500 MB	Original + subtitled MP4 + SRT/JSON
jobs/ — video de 1h, 4 1 language	~2–4 GB	One subtitled MP4 per language
jobs/ — audio of 1h	~100–200 MB	MP3 + text/subtitle files
jobs/ — PDF document	~5–20 MB	PDF + translated files

- The `jobs/` folder will grow indefinitely if a retention policy is not implemented. It is recommended to delete tasks older than 30 days..
- Recommended total storage: 100–200 GB for moderate use (up to 50 one-hour videos with 4 languages).

3. Network

Aiuda processes all files locally once the models have been downloaded. It does not send data to external servers during processing.

Concept	Details
Initial installation	Model download: ~3 GB. Requires internet access (Hugging Face and PyPI).
Processing	100% offline. No data is sent to external servers.
File upload	A 1-hour video can be 1–4 GB in size. On a 1 Gbps LAN, the transfer is instantaneous.
Institutional proxy	If the server is behind a proxy, NO_PROXY must be configured for internal communication between Docker containers.

4. Estimated cost

The costs below are reference estimates in US dollars for server hardware. If infrastructure is already available, the additional cost may be zero.

		Costo aprox. (USD)
Physical server (used)	Dell/HP 2U, 24 cores, 64 GB RAM	$800–$2.000
Physical server (new)	Tower workstation, 16 cores, 32 GB RAM	$2.500–$5.000
GPU (optional)	NVIDIA RTX 3060/4060 (12 GB VRAM)	$300–$500
SSD storage	500 GB NVMe para jobs/	$40–$80
Software	Linux + Docker + Aiuda	$0 (libre)
Total without GPU	Used Server + SSD	~$850–$2.100
Total with GPU	Server + GPU + SSD	~$1.150–$2.600

- If the institution already has a server with available capacity, the additional cost is limited to storage (~$40–$80) and, optionally, a GPU.
- Prices may vary depending on the region and market availability.

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